



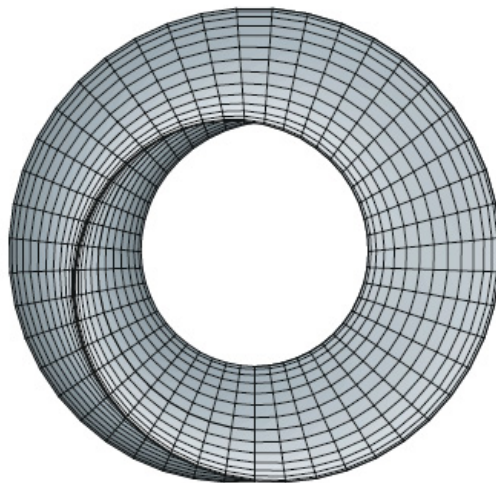
UNIVERSITY OF ZIMBABWE

HMTHCS111 CALCULUS 2

Calculus of Several Variables

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August 28, 2026

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Chapter 3

Lines and Planes in Space

Gravity explains the motions of the planets, but it cannot explain who sets the planets in motion.

— Isaac Newton

In the plane, **slope** is used to determine an equation of a line. In space, it is more convenient to use **vectors** to determine the equation of a line.

3.1 Vector Equation of a Straight Line

Just as in finding the equation of a straight line in Cartesian coordinates, we will find that there are two cases to consider when finding the equation of a straight line in terms of vectors.

Case I:

One case is when you are given a fixed point on the straight line and a vector parallel to the straight line. This case is analogous to the situation in Cartesian coordinates when one is given a point through which a straight line passes and the gradient of the line is known.

Suppose the straight line l passes through a point A with position vector $\mathbf{a}$ with respect to our reference point O . Further, suppose the vector $\mathbf{b}$ is parallel to the straight line. Let R be any point on the straight line and let its position vector relative to O be $\mathbf{r}$. We have

$$\overrightarrow{OR} = \overrightarrow{OA} + \overrightarrow{AR}.$$

Since $\overrightarrow{AR}$ is parallel to the vector $\mathbf{b}$, $\overrightarrow{AR} = \lambda \mathbf{b}$ where λ is a real number. Hence the vector equation of a straight line is given by

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}. \tag{3.1}$$

Case II:

Suppose you are given two points through which the straight line passes.

Suppose a straight line l passes through two points A and B with position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively relative to the reference point O . Let R be any point on the straight line and let its position vector relative to O be $\mathbf{r}$. We have

$$\overrightarrow{OR} = \overrightarrow{OA} + \overrightarrow{AR}.$$

But $\overrightarrow{AR} = \lambda \overrightarrow{AB}$ where $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$.

Thus $\mathbf{r} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a}) = (\mathbf{a} - \lambda\mathbf{a}) + \lambda\mathbf{b} = (1 - \lambda)\mathbf{a} + \lambda\mathbf{b}$. Therefore the vector equation of a straight line in this case is given by

$$\mathbf{r} = (1 - \lambda)\mathbf{a} + \lambda\mathbf{b}. \quad (3.2)$$

3.2 The Cartesian Equation of a Straight Line

We know that the general equation of a straight line in 2 dimensions has the form $ax + by + c = 0$ or the equivalent form $y = mx + c$.

You might expect the general equation of a straight line in 3 dimensional space to have the form $ax + by + cz + d = 0$. In fact this is the general equation of a **plane** and **not** a straight line. In general, there are two Cartesian coordinate forms of the equation of a straight line in 3 dimensional space namely:

1. parametric equations form
2. symmetric equations form.

They are all derived from the vector equation (3.1).

Parametric Equations

Suppose vectors $\mathbf{r}$, $\mathbf{a}$, $\mathbf{b}$ have column vectors $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$, $\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$ respectively and suppose $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ is the vector equation of straight line l .

Then $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + \lambda \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$.

Thus, we have

$$x = x_1 + \lambda x_2, \quad y = y_1 + \lambda y_2, \quad z = z_1 + \lambda z_2 \quad (3.3)$$

These are the parametric equations of the straight line l with parameter λ in 3 dimensional space.

Symmetric Equations

From the equations in (3.3), we obtain

$$\frac{x - x_1}{x_2} = \frac{y - y_1}{y_2} = \frac{z - z_1}{z_2} = \lambda.$$

Hence, we have

$$\frac{x - x_1}{x_2} = \frac{y - y_1}{y_2} = \frac{z - z_1}{z_2}. \quad (3.4)$$

These are the symmetric form of the Cartesian equations of a straight line in 3 dimensional space.

Consider the line L through the point $P = (x_1, y_1, z_1)$ and parallel to vector $\mathbf{A} = (a, b, c)$. The vector $\mathbf{A}$ is the **direction vector** for the line L , and a, b, c are the **direction numbers**. One way of describing the line L is to say that it consists of all points $Q = (x, y, z)$ for which the vector $\overrightarrow{PQ}$ is parallel to $\mathbf{A}$. This means that $\overrightarrow{PQ}$ is a scalar multiple of $\mathbf{A}$, and you can write $\overrightarrow{PQ} = t\mathbf{A}$, where t is a scalar.

$$\overrightarrow{PQ} = (x - x_1, y - y_1, z - z_1) = (at, bt, ct) = t\mathbf{A}.$$

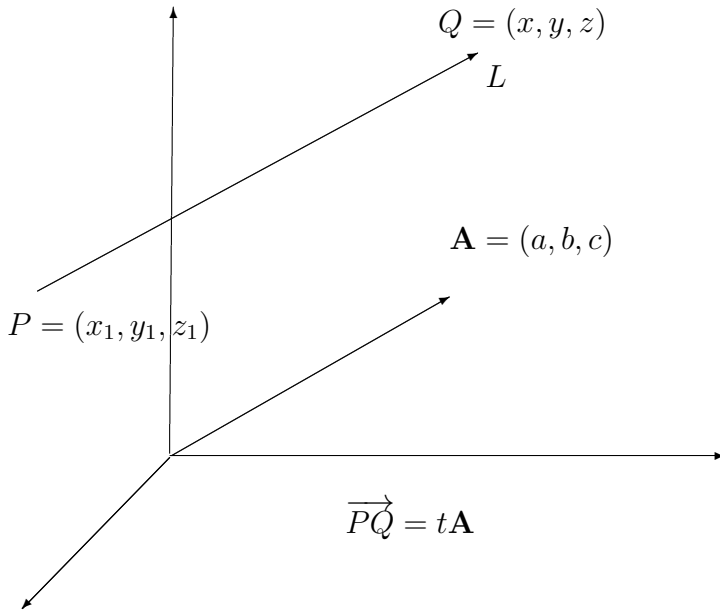


Figure 3.1: A line has directed vector $\mathbf{A}$

3.3 Parametric and Symmetric Equations of a Line in Space

By equating corresponding components, you can obtain the **parametric equations** of a line in space.

Theorem 3.3.1. *A line L parallel to the vector $\mathbf{A} = (a, b, c)$ and passing through the point $P = (x_1, y_1, z_1)$ is represented by the **parametric equations***

$$x = x_1 + at, \quad y = y_1 + bt, \quad z = z_1 + ct.$$

3.3.1 Symmetric Equations

If the direction numbers a, b, c are all non-zero, then you can eliminate the parameter t to obtain the **symmetric equations** of a line.

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}.$$

Example 3.3.1. *Find parametric and symmetric equations of the line L that passes through the point $(1, -2, 4)$ and is parallel to $\mathbf{A} = (2, 4, -4)$.*

Solution: *To find a set of parametric equations of the line, use the coordinates $x_1 = 1, y_1 = -2$ and $z_1 = 4$ and the direction numbers $x_2 = 2, y_2 = 4$ and $z_2 = -4$.*

$$x = 1 + 2\lambda, \quad y = -2 + 4\lambda, \quad z = 4 - 4\lambda.$$

Because x_2, y_2 and z_2 are all non-zero, a set of symmetric equations is

$$\frac{x - 1}{2} = \frac{y + 2}{4} = \frac{z - 4}{-4}.$$

Neither the parametric equations nor the symmetric equations of a given line are unique.

Example 3.3.2. *Find a set of parametric equations of the line that passes through the points $(-2, 1, 0)$ and $(1, 3, 5)$.*

Solution: *Begin by letting $P = (-2, 1, 0)$ and $Q = (1, 3, 5)$. Then a direction vector for the line passing through P and Q is given by*

$$\mathbf{A} = (1 - (-2), 3 - 1, 5 - 0) = (3, 2, 5) = (x_2, y_2, z_2).$$

Using the direction numbers $x_2 = 3, y_2 = 2$ and $z_2 = 5$, with the point $P = (-2, 1, 0)$ you can obtain the parametric equations

$$x = -2 + 3\lambda, \quad y = 1 + 2\lambda, \quad z = 5\lambda.$$

3.4 Planes in Space

We have seen how an equation of a line in space can be obtained from a point on the line and a vector *parallel* to it. Now we will see that an equation of a plane in space can be obtained from a point in the plane and a vector *normal* (perpendicular) to it.

Consider the plane containing the point $P = (x_1, y_1, z_1)$ and having a non-zero normal vector $\mathbf{n} = (a, b, c)$. This plane consists of all points $Q = (x, y, z)$ for which vector $\overrightarrow{PQ}$ is orthogonal to $\mathbf{n}$. Using the dot product, we have the following

$$\begin{aligned}\mathbf{n} \cdot \overrightarrow{PQ} &= 0 \\ (a, b, c) \cdot (x - x_1, y - y_1, z - z_1) &= 0 \\ a(x - x_1) + b(y - y_1) + c(z - z_1) &= 0.\end{aligned}$$

The third equation of the plane is said to be in **standard form**.

3.4.1 Standard Equation of a Plane in Space

Theorem 3.4.1. *The plane containing the point (x_1, y_1, z_1) and having a normal vector $\mathbf{n} = (a, b, c)$ can be represented, in **standard form** by the equation*

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0.$$

By regrouping terms, we obtain the **general form** of the equation of a plane in space,

$$ax + by + cz + d = 0.$$

Given the general form of the equation of a plane, it is easy to find a normal vector to the plane. simply use the coefficients of x, y and z and write $\mathbf{n} = (a, b, c)$.

Example 3.4.1. *Find the general equation of the plane containing the points $(2, 1, 1)$, $(0, 4, 1)$ and $(-2, 1, 4)$.*

Solution: *We need a point in the plane and a vector that is normal to the plane. There are three choices for the point, but no normal vector is given. To obtain a normal vector, use the cross product of vectors $\mathbf{B}$ and $\mathbf{C}$ extending from the point $(2, 1, 1)$ to the points $(0, 4, 1)$ and $(-2, 1, 4)$. The component forms of $\mathbf{B}$ and $\mathbf{C}$ are*

$$\begin{aligned}\mathbf{B} &= (0 - 2, 4 - 1, 1 - 1) = (-2, 3, 0) \\ \mathbf{C} &= (-2 - 2, 1 - 1, 4 - 1) = (-4, 0, 3)\end{aligned}$$

and it follows that

$$\mathbf{n} = \mathbf{B} \times \mathbf{C} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 3 & 0 \\ -4 & 0 & 3 \end{vmatrix} = 9\mathbf{i} + 6\mathbf{j} + 12\mathbf{k} = (a, b, c)$$

is normal to the given plane. Using the direction numbers for $\mathbf{n}$ and the point $(x_1, y_1, z_1) = (2, 1, 1)$, you can determine an equation of the plane to be

$$\begin{aligned} a(x - x_1) + b(y - y_1) + c(z - z_1) &= 0 \\ 9(x - 2) + 6(y - 1) + 12(z - 1) &= 0 \quad (\text{Standard form}) \\ 9x + 6y + 12z - 36 &= 0 \\ 3x + 2y + 4z - 12 &= 0. \quad (\text{General form}) \end{aligned}$$

Remark: Check to see that each of the three points satisfies the equation $3x + 2y + 4z - 12 = 0$.

3.5 Angle Between Two Planes

Two distinct planes in three-dimensional space either are parallel or intersect in a line. If they intersect, you can determine the angle between them from the angle between their normal vectors. Specifically, if vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ are normal to two intersecting planes, the angle θ between the normal vectors is equal to the angle between the two planes and is given by

$$\cos \theta = \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1||\mathbf{n}_2|}.$$

Consequently, two planes with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ are

1. *perpendicular*, if $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$.
2. *parallel*, if $\mathbf{n}_1$ is a scalar multiple of $\mathbf{n}_2$.

Example 3.5.1. Find the angle between the two planes

$$\begin{aligned} x - 2y + z &= 0 && \text{Equation for Plane 1} \\ 2x + 3y - 2z &= 0 && \text{Equation for Plane 2} \end{aligned}$$

and find parametric equations of their line of intersection.

Solution: The normal vectors for the planes are $\mathbf{n}_1 = (1, -2, 1)$ and $\mathbf{n}_2 = (2, 3, -2)$. Consequently, the angle between the two planes is as follows

$$\begin{aligned} \cos \theta &= \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1||\mathbf{n}_2|} \\ &= \frac{-6}{\sqrt{6}\sqrt{17}} \\ &= \frac{-6}{\sqrt{102}} \\ &\approx -0.59409 \quad (\theta \approx \cos^{-1} -0.59409). \end{aligned}$$

You can find the line of intersection of the two planes by simultaneously solving the two linear equations representing the planes. Multiply the first equation by -2 and add the result to the second equation.

$$\begin{array}{r} -2x + 4y - 2z = 0 \\ \underline{2x + 3y - 2z = 0} \\ 0x + 7y - 4z = 0. \end{array}$$

So $y = \frac{4z}{7}$. Substituting this into one of the original equations, we have $x = \frac{z}{7}$. Finally, letting $t = \frac{z}{7}$, we obtain the parametric equations

$$x = t, \quad y = 4t, \quad z = 7t \quad (\text{Line of Intersection}).$$

3.6 Distances Between Points, Planes and Lines

In this section we want to find two basic types of distance problems in space,

1. Finding the distance between a point and a plane.
2. Finding the distance between a point and a line.

The solutions of these problems illustrate the versatility and usefulness of vectors in coordinate geometry. The first problem uses the *dot product* of two vectors, and the second problem uses the *cross product*. The distance D between a point Q and a plane is the length of the shortest line segment connecting Q to the plane.

3.6.1 Distance Between a Point and a Plane

Theorem 3.6.1. *The distance between a plane and a point Q (not in the plane) is*

$$D = \frac{|\overrightarrow{PQ} \cdot \mathbf{n}|}{|\mathbf{n}|}.$$

To find a point in the plane given by $ax + by + cz + d = 0$ ($a \neq 0$), let $y = 0$ and $z = 0$. Then, from the equation $ax + d = 0$, we conclude that the point $(-\frac{d}{a}, 0, 0)$ lies in the plane.

Example 3.6.1. *Find the distance between $Q = (1, 5, -4)$ and the plane given by $3x - y + 2z = 6$.*

Solution: We know that $\mathbf{n} = (3, -1, 2)$ is normal to the given plane. To find a point in the plane, let $y = 0$ and $z = 0$, and obtain the point $P = (2, 0, 0)$. The vector from P to Q is given by

$$\overrightarrow{PQ} = (1 - 2, 5 - 0, -4 - 0) = (-1, 5, -4).$$

Using the distance formula,

$$\begin{aligned} D &= \frac{|\overrightarrow{PQ} \cdot \mathbf{n}|}{|\mathbf{n}|} = \frac{|(-1, 5, -4) \cdot (3, -1, 2)|}{\sqrt{9 + 1 + 4}} \\ &= \frac{|-3 - 5 - 8|}{\sqrt{14}} \\ &= \frac{16}{\sqrt{14}}. \end{aligned}$$

Remark: The choice of the point P is arbitrary in the Example. Try choosing a different point to verify that we obtain the same distance.

The distance between the point $Q = (x_0, y_0, z_0)$ and the plane given by $ax + by + cz + d = 0$ is

$$\begin{aligned} D &= \frac{|a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1)|}{\sqrt{a^2 + b^2 + c^2}} \\ D &= \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}} \end{aligned}$$

where $P = (x_1, y_1, z_1)$ is a point on the plane and $d = -(ax_1 + by_1 + cz_1)$.

3.7 Finding the Distance Between Two Parallel Planes

Example 3.7.1. Find the distance between the two parallel planes given by $3x - y + 2z - 6 = 0$ and $6x - 2y + 4z + 4 = 0$.

Solution: To find the distance between the planes, choose a point in the first plane, say $(x_0, y_0, z_0) = (2, 0, 0)$. Then, from the second plane, we determine that $a = 6, b = -2, c = 4$ and $d = 4$, and conclude that the distance is

$$\begin{aligned} D &= \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{|6(2) + (-2)(0) + (4)(0) + 4|}{\sqrt{6^2 + (-2)^2 + 4^2}} \\ &= \frac{16}{\sqrt{56}} = \frac{8}{\sqrt{14}} \approx 2.14. \end{aligned}$$

3.8 Distance Between a Point and a Line in Space

The formula for the distance between a point and a line in space resembles that for the distance between a point and a plane, except that we replace the dot product by the cross product and replace the normal vector $\mathbf{n}$ by a direction vector for the given line.

Theorem 3.8.1. *The distance between a point Q and a line in space is given by*

$$D = \frac{|\overrightarrow{PQ} \times \mathbf{u}|}{|\mathbf{u}|}$$

where $\mathbf{u}$ is the direction vector for the line and P is a point on the line.

Example 3.8.1. *Find the distance between the point $Q = (3, -1, 4)$ and the line given by*

$$x = -2 + 3t, \quad y = -2t, \quad z = 1 + 4t.$$

Solution: *Using the direction numbers, 3, -2 and 4, we know that the direction vector for the line is $\mathbf{u} = (3, -2, 4)$. To find a point on the line, let $t = 0$, and obtain $P = (-2, 0, 1)$. thus,*

$$\overrightarrow{PQ} = (3 - (-2), -1 - 0, 4 - 1) = (5, -1, 3)$$

and we form the cross product

$$\overrightarrow{PQ} \times \mathbf{u} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5 & -1 & 3 \\ 3 & -2 & 4 \end{vmatrix} = 2\mathbf{i} - 11\mathbf{j} - 7\mathbf{k} = (2, -11, -7).$$

Finally

$$D = \frac{|\overrightarrow{PQ} \times \mathbf{u}|}{|\mathbf{u}|} = \frac{\sqrt{174}}{\sqrt{29}} = \sqrt{6} \approx 2.45.$$

3.9 Tutorial Questions

- Find the parametric and symmetric equations of the line through the two points.
(i) $(5, -3, -2)$ and $(-\frac{2}{3}, \frac{2}{3}, 1)$ (ii) $(1, 0, 1)$ and $(1, 3, -2)$.
- Determine which of the points lie on the line L .
(a) The line L passes through the point $(-2, 3, 1)$ and is parallel to the vector $\mathbf{A} = 4\mathbf{i} - \mathbf{k}$.
(i) $(2, 3, 0)$ (ii) $(-6, 3, 2)$ (iii) $(2, 1, 0)$ (iv) $(6, 3, -2)$.
(b) The line L passes through the points $(2, 0, -3)$ and $(4, 2, -2)$.
(i) $(4, 1, -2)$ (ii) $(\frac{5}{2}, \frac{1}{2}, -\frac{11}{4})$ (iii) $(-1, -3, -4)$ (iv) $(0, -2, -4)$.
- Find the parametric equation of the line that passes through the point $(2, 3, 4)$ and is perpendicular to the plane given by $3x + 2y - z = 6$.

4. Find the equation of the plane that contains the point $(1, -2, 3)$ and is parallel to the plane $z = 2x + 3y - 4$.
5. Find the equation of the plane passing through the points
 - (i) $(0, 0, 0)$, $(1, 2, 3)$ and $(-2, 3, 3)$
 - (ii) $(1, 2, -3)$, $(2, 3, 1)$ and $(0, -2, -1)$
 - (iii) $(1, 2, 3)$, $(3, 2, 1)$ and $(-1, -2, 2)$.
6. Determine whether the planes are parallel or orthogonal or neither. If they are neither parallel nor orthogonal, find the angle between them.
 - (i) $5x - 3y + z = 4$, $x + 4y + 7z = 1$
 - (ii) $3x + y - 4z = 3$, $-9x - 3y + 12z = 4$
 - (iii) $x - 3y + 6z = 4$, $5x + y - z = 4$
 - (iv) $3x + 2y - z = 7$, $x - 4y + 2z = 0$
 - (v) $x - 5y - z = 1$, $5x - 25y - 5z = -3$
 - (vi) $2x - z = 1$, $4x + y + 8z = 10$.
7. Find the angle of intersection of the plane $x + y - z = 0$ and the plane $x - 3y + z - 1 = 0$.
8. Given the planes $3x + y - 5 = 0$, $y + z - 1 = 0$. Determine equations of the line of intersection of the planes.
9. Find the parametric equations for the line of intersection of the planes.
 - (i) $3x + 2y - z = 7$, $x - 4y + 2z = 0$
 - (ii) $x - 3y + 6z = 4$, $5x + y - z = 4$
 - (iii) $2x + y + z = 4$, $3x - y + z = 3$.
10. Find the parametric equations of the line of intersection of the planes $2x + y + z = 4$ and $3x - y + z = 3$.
11. Find the distance between the point $(10, 3, -2)$ and the line $x = 4t - 2$, $y = 3$, $z = -t + 1$.
12. Find the distance between the point $(4, 1, -2)$ and the line $x = 2t + 2$, $y = 2t$, $z = t - 3$.
13. Find the distance between the point $(0, 0, 0)$ and the plane $2x + 3y + z = 12$.
14. Find the distance from the point $(2, 5, 4)$ to the plane $x + 2y + 2z = 2$.
15. Find the distance between the point $(1, 2, 3)$ and the plane $2x - y + z = 4$.
16. Find the distance between the planes.
 - (i) $x - 3y + 4z = 10$, $x - 3y + 4z = 6$
 - (ii) $2x - 4z = 4$, $2x - 4z = 10$
 - (iii) $2x - 3y + z = 6$, $2x - 3y + z = 2$
 - (iv) $x + 2y - z = 0$, $x + 2y - z = 4$.
17. A plane passes through the point $(1, 1, 1)$ and is perpendicular to each of the planes $3x - 2y + 3z + 6 = 0$ and $6x - 2y - 3z - 6 = 0$. Find its equation.
18. Given three points $(1, 2, 3)$, $(-1, 3, 1)$ and $(-2, -3, 0)$, find the equation of the plane through these points.
19. Find an equation for the plane that passes through the point $(1, 3, -2)$ and also passes through the line of intersection of the planes $x - y + z = 1$ and $x + y - z = 1$.
20. Find the equation of the plane through the points $(1, 0, -1)$ and $(2, 1, 0)$ that is also parallel to the line of intersection of the planes $x + y + z = 5$ and $3x - y = 4$.